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## AWS Systems Manager (SSM) – Overview

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### What is AWS Systems Manager?

**AWS Systems Manager** is a **unified operations hub** that lets you **automate, manage, and monitor** your AWS resources **securely and at scale**.

It helps DevOps, SRE, and Cloud Engineers manage infrastructure **without logging in to servers**.

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### Key Capabilities

| Feature             | Description                                         |
|---------------------|-----------------------------------------------------|
| Session Manager     | Secure shell access to EC2 (no SSH, no open ports)  |
| Run Command         | Remotely execute scripts/commands on EC2 or on-prem |
| Automation          | Orchestrate tasks (e.g., patching, backups)         |
| Patch Manager       | Automatically patch OS updates                      |
| Inventory           | Collect software & configuration details            |
| Parameter Store     | Securely store config variables and secrets         |
| OpsCenter           | Central console for managing operational issues     |
| State Manager       | Enforce instance configurations over time           |
| Maintenance Windows | Schedule actions on EC2 (e.g., patches, updates)    |

## Compliance Dashboard

Track patch compliance and config drift

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### Why Use It?

-  **No SSH needed** – use **Session Manager**
  -  **Secure** – integrates with IAM and KMS
  -  **Centralized automation** for multi-instance workflows
  -  **Audit-ready** – actions are logged to **CloudTrail**
  -  **Works with EC2, on-prem, and hybrid environments**
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### SSM Agent

- **SSM Agent** must be installed and running on managed instances.
  - Pre-installed on:
    - Amazon Linux 2
    - Ubuntu 16.04+, Windows 2012+
  - You need to **attach IAM role** with [AmazonSSMManagedInstanceCore](#) policy to EC2.
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## Example Use Cases

| Task                                 | SSM Tool                              |
|--------------------------------------|---------------------------------------|
| Secure shell access without key pair | <b>Session Manager</b>                |
| Install software on 100+ servers     | <b>Run Command</b>                    |
| Enforce NTP or antivirus on boot     | <b>State Manager</b>                  |
| Patch Windows every Sunday night     | <b>Maintenance Window</b>             |
| Store DB password securely           | <b>Parameter Store (SecureString)</b> |

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## Hands-On Demo Ideas

### 1. Run a command on EC2 instance

```
aws ssm send-command \  
  --document-name "AWS-RunShellScript" \  
  --targets "Key=instanceIds,Values=i-xxxxxxxxxxxx" \  
  --comment "Check disk usage" \  
  --parameters commands="df -h" \  
  --region us-east-1
```

### 2. Open EC2 instance via Session Manager

- Go to: **AWS Console** → **Systems Manager** → **Session Manager**
- Start session with any managed instance (with SSM agent + role)

### 3. Store a secret in Parameter Store

```
aws ssm put-parameter \  
  --name "/dev/db/password" \  
  --value "mySecretPassword123" \  
  --type "SecureString"
```

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#### Pricing

##### Free for most features

 Some features (like Automation executions or inventory collection) may incur minimal costs per use or per instance.

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#### Summary

| Feature   | Description                                             |
|-----------|---------------------------------------------------------|
| What      | Manage AWS infrastructure at scale                      |
| How       | Use tools like Run Command, Session Manager, Automation |
| Use cases | Secure access, patching, config mgmt, secret storage    |
| Support   | EC2, on-prem, hybrid                                    |
| Secure    | IAM + KMS + no SSH needed                               |

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